Semana 7: A Exponencial Complexa

MOTIVAÇÃO: Queremos definir a exponencial complexa de modo que ela seja uma extensão da exponencial real e, portanto, preserve suas propriedades

$$e^z = e^{x+iy} = e^x e^{iy}$$

Série de McLaurin

$$e^{iy} = \sum_{n=0}^{\infty} \frac{(iy)^n}{n!} = 1 + i\frac{y}{1!} - \frac{y^2}{2!} - i\frac{y^3}{3!} + \frac{y^4}{4!} + i\frac{y^5}{5!} + \cdots = \left(1 - \frac{y^2}{2!} + \frac{y^4}{4!} + \cdots\right) + i\left(\frac{y}{1!} - \frac{y^3}{3!} + i\frac{y^5}{5!} + \cdots\right)$$

$$= \sum_{n=0}^{\infty} (-1)^n \frac{y^{2n}}{(2n)!} + i\sum_{n=0}^{\infty} (-1)^n \frac{y^{2n+1}}{(2n+1)!} = \cos y + i \sin y$$

$$e^z = e^{x+iy} = e^x(\cos y + i \sin y)$$

$$e^{z_1} e^{z_2} = e^{z_1+z_2}$$

$$e^{z_1} e^{z_2} = e^{x_1}(\cos y_1 + i \sin y_1) e^{x_2}(\cos y_2 + i \sin y_2)$$
$$= e^{x_1+x_2}[(\cos y_1 \cos y_2 - \sin y_1 \sin y_2) + i(\sin y_1 \cos y_2 + \cos y_1 \sin y_2)]$$
$$= e^{x_1+x_2}(\cos(y_1 + y_2) + i \sin(y_1 + y_2)) = e^{(x_1+x_2)+i(y_1+y_2)} = e^{z_1+z_2}$$

$$e^{-z} = \frac{1}{e^z}$$

$$e^{-z} = e^{-x-iy} = e^{-x}(\cos(-y) + i \sin(-y)) = \frac{1}{e^x}(\cos y - i \sin y) = \frac{1}{e^x(\cos y + i \sin y)} = \frac{1}{e^{x+iy}} = \frac{1}{e^z}$$

$$\frac{1}{z} = \frac{\bar{z}}{|z|^2}$$

$$(e^z)^n = e^{nz}, \quad n \in \mathbb{Z}$$

Indução: $(n \in \mathbb{N})$

$n = 0$: $\quad (e^z)^0 = 1 = e^{0z}$

$n \Longrightarrow n + 1$:

$$(e^z)^{n+1} = (e^z)^n e^z = e^{nz} e^z = e^{nx+iny}\, e^{x+iy} = e^{nx}\,(\cos(ny) + i\sin(ny))e^x(\cos y + i\sin y)$$

$$= e^{(n+1)x}[(\cos(ny)\cos y - \sin(ny)\sin y) + i(\sin(ny)\cos y + \cos(ny)\sin y)]$$

$$= e^{(n+1)x}\big(\cos\big((n+1)y\big) + i\sin\big((n+1)y\big)\big) = e^{(n+1)x+i(n+1)y} = e^{(n+1)z}$$

$n < 0$:

$$e^{nz} = e^{-mz} = \frac{1}{e^{mz}} = \frac{1}{(e^z)^m} = (e^z)^{-m} = (e^z)^n$$

A exponencial complexa é uma extensão da extensão real:

$$e^{x+i0} = e^{x}(\cos 0 + i \sin 0) = e^{x}$$

Representação Polar:

$$z = r(\cos\theta + i \sin\theta) = re^{i\theta}$$

Fórmula de De Moivre: $\left(e^{i\theta}\right)^{n} = e^{in\theta}$

Outra Notação: $e^{z} \equiv \exp(z)$

Exemplo 1: $\boldsymbol{e^{2\pi i} = \cos(2\pi) + i\sin(2\pi) = 1}$

Exemplo 2: $z = -\sqrt{3} + i$

$$r = |z| = \sqrt{\left(-\sqrt{3}\right)^2 + 1^2} = \sqrt{4} = 2$$

$$\begin{cases} \cos\theta = \dfrac{Re(z)}{|z|} = \dfrac{-\sqrt{3}}{2} \\ \sin\theta = \dfrac{Im(z)}{|z|} = \dfrac{1}{2} \end{cases} \Longrightarrow \theta = \frac{5\pi}{6}$$

$$z = 2e^{i\frac{5\pi}{6}}$$